

Crizotinib in *MET* deregulated or *ROS1* rearranged pretreated non-small-cell lung cancer (METROS): a phase II, prospective, multicentre, two-arms trial

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Statement of translational relevance

Treatment of patients with *MET* deregulated Non-Small Cell Lung Cancer (NSCLC) represents an urgent need because of lack of effective targeted therapies and unfavourable prognosis. The METROS trial is a prospective study evaluating the efficacy of crizotinib in two cohorts of patients: individuals with *MET* exon 14 mutations or amplification or individuals with *ROS1* rearrangements. In the *MET* deregulated cohort, although response rate was 27%, a remarkable result for a pretreated NSCLC population, median PFS and most importantly, median OS were disappointing with all patients rapidly progressing and dying. Interestingly, no difference for any clinical end-point was observed between *MET* amplified and exon 14 mutated patients. Overall, our data highlight the urgent need for more effective strategies in patients with *MET* mutations or amplification.

104 **Abstract**

105 **Introduction**

106 *MET* deregulated NSCLC represents an urgent clinical need because of unfavorable
107 prognosis and lack of specific therapies. Although recent studies have suggested a
108 potential role for crizotinib in patients harboring *MET* amplification or exon 14 mutations,
109 no conclusive data are currently available. The present study aimed at investigating
110 activity of crizotinib in patients harboring *MET* or *ROS1* alterations.

111 **Methods**

112 Patients with pretreated advanced NSCLC and evidence of *ROS1* rearrangements (Cohort
113 A) or *MET* deregulation (amplification, ratio $MET/CEP7 > 2.2$ or *MET* exon 14 mutations,
114 Cohort B) were treated with crizotinib 250 mg BID orally. The co-primary end-point was
115 objective response rate in the two cohorts.

116 **Results**

117 From December 2014 to March 2017, 505 patients were screened and a total of 52
118 patients (26 patients per cohort) were enrolled onto the study. At data cut-off of September
119 2017, in cohort A objective response rate was 65%, median progression free survival and
120 overall survival were 22.8 months (95% CI 15.2 – 30.3) and not reached. In cohort B,
121 objective response rate was 27%, median progression free survival was 4.4 months (95%
122 CI 3.0- 5.8) and overall survival was 5.4 months (95% CI 4.2– 6.5). No difference in any
123 clinical end-point was observed between *MET* amplified and exon14 mutated patients. No
124 response was observed among the 5 patients with co-occurrence of a second gene
125 alteration. No unexpected toxicity was observed in both cohorts.

126 **Conclusions**

127 Crizotinib induces response in a fraction of *MET* deregulated NSCLC. Additional studies
128 and innovative therapies are urgently needed.

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133 Introduction

134 During the last ten years several molecular events, including gene mutations, gene copy
135 number alterations and gene rearrangements have been discovered in small fractions of
136 lung adenocarcinomas, dramatically improving patient treatment (1). This is the case of
137 NSCLCs carrying epidermal growth factor receptor (*EGFR*) activating mutations or
138 anaplastic lymphoma kinase (*ALK*) rearrangement, where targeted therapies have
139 changed the natural history of the disease (2–5). Beyond *EGFR* mutations and *ALK*
140 rearrangement, additional actionable alterations have been identified, with *ROS1*
141 rearrangement and *MET* amplification or *MET* exon 14 mutations being the most
142 appealing (5–8).

143 Crizotinib has been the first *ALK* inhibitor entered in clinical development within the
144 PROFILE program and then approved worldwide for front line treatment of *ALK* positive
145 NSCLC (3). Moreover, crizotinib has high specificity for *ROS1* and *MET* kinase domains
146 and it potently inhibits tumor growth and invasion in cell models of *MET* or *ROS1* addicted
147 NSCLC (9). In the phase I PROFILE 1001, enrolling 50 *ROS1*-positive NSCLC patients,
148 objective response rate (ORR) was 72% and median progression-free survival (PFS)
149 exceeded 19 months (10). These results mirror those observed in *ALK* positive NSCLC
150 and are comparable to what reported in subsequent retrospective and prospective trials
151 (11–15). At present, crizotinib has a well established role in *ROS1* positive NSCLC and is
152 worldwide available. Conversely, its role in *MET* addicted NSCLC is not demonstrated.

153 Initial observations in solid tumors with *MET* overexpression or *MET* amplification showed
154 potential efficacy only in *de novo MET*-amplified tumors (16–18). In NSCLC, crizotinib
155 produced an ORR of approximately 40%, with evidence of activity only against tumors with
156 intermediate or high levels of *MET* amplification, defined as a ratio *MET*/centromere 7
157 (*MET*/CEP7) of $> 2.2 - < 5$ or ≥ 5 , respectively (19). Interestingly, preliminary results of the
158 phase II AcSé trial, evaluating crizotinib in *MET* amplified NSCLC, showed an ORR of only

32% (20). Nevertheless, the criteria adopted for defining *MET* amplification were different than in the PROFILE 1001 study, providing a possible explication for the lower drug efficacy. Moreover, AcSé data suggested that levels of *MET* amplification could be relevant for defining patients with the highest sensitivity to the drug. In addition to *MET* amplification, a recent study showed that anti-MET agents such as crizotinib or cabozantinib induced tumor shrinkage in patients harboring *MET* exon 14 mutations, a phenomenon occurring in approximately 3% of NSCLC (7,8,21,22). In the phase 1 PROFILE 1001 study, ORR with crizotinib in *MET* exon 14 mutated patients was 44%, suggesting that the drug is effective against this alteration (23). Based on these premises and considering the urgent need of effective strategies for *MET* deregulated NSCLC, we designed this study aiming at investigating crizotinib efficacy in *MET* amplified or exon 14 mutated NSCLC. Because at the time of study design, few data were available in *ROS1* rearranged patients and no therapy was available in Italian clinical practice, a *ROS1* rearranged cohort was also included.

Methods

Patients selection

Eligible patients had histologically confirmed diagnosis of locally advanced or metastatic NSCLC and availability of archival tissue for biomarkers analyses. A local pre-screening was allowed. For patients with *MET* exon 14 mutations central confirmation was not required for trial inclusion, whereas central confirmation was mandatory for those patients who resulted positive for *ROS1* rearrangements or *MET* amplification at local labs. *ROS1* rearrangement and *MET* amplification were tested centrally by fluorescent in-situ hybridization (FISH), using the specific probes (Abbott Molecular, Des Plaines, IL, USA). Briefly, criteria for FISH positivity were 1) presence of *ROS1*-fusion patterns in $\geq 15\%$ of tumor cells (24) or 2) a *MET*/CEP7 ratio > 2.2 according to Camidge Criteria (19). *MET*

185 mutational status was assessed locally using direct sequencing or other high sensitive
186 methods. Even if central confirmation was not required for trial inclusion, at the end of the
187 study, all *MET* exon 14 mutant cases were centrally verified using Sanger direct
188 sequencing (Applied Biosystem - ThermoFisher Scientific, San Francisco, CA, USA).
189 Other inclusion criteria included: an Eastern Cooperative Oncology Group performance
190 status (ECOG PS) ≤ 2 , at least one previous chemotherapy line, at least one measurable
191 tumor lesion according to Response Evaluation Criteria in Solid Tumors (RECIST) version
192 1.1 (25), adequate bone marrow and organ functions. Patients with known *EGFR* or *KRAS*
193 mutations or previously treated with ROS1 or MET inhibitors or with symptomatic brain
194 metastases were excluded. The study was done in accordance with the provisions of the
195 Declaration of Helsinki and Good Clinical Practice guidelines. Each center received the
196 approval of the local ethics committee, and all patients provided written informed consent
197 before participation. The final version of the protocol including full list of study criteria is
198 reported in the Supplemental data.

199 *Treatment*

200 Patients were treated with crizotinib 250 mg twice daily in continuous 28-day cycles until
201 disease progression, unacceptable toxicity, withdrawal of consent, or death. Dose
202 modifications or interruptions were considered in case of intolerable grade 2 or worse
203 adverse events (AEs). Radiological assessment by CT scans was done at baseline and
204 then every 8 weeks until disease progression; responses had to be confirmed by repeating
205 assessment 4–8 weeks after initial response. All patients who discontinued study drug
206 were followed up for subsequent treatments and survival every 12 weeks, until death or
207 study completion. Patients were assessed for safety every 4 weeks. AEs, laboratory tests,
208 and vital signs were graded according to the Common Terminology Criteria for Adverse
209 Events version 4.0. The cutoff date for safety and efficacy data was September 30, 2017,
210 which was the date of database lock.

211 Outcomes

212 The primary endpoint was investigator-assessed overall response, defined as the
213 percentage of patients who achieved a confirmed complete response (CR) or partial
214 response (PR) per RECIST version 1.1 (25). Secondary endpoints included PFS based on
215 investigator-assessed disease response OS, safety and correlation between response and
216 percentage of *ROS1* FISH positivity or levels of *MET* amplification (intermediate levels,
217 [MET/CEP7 ratio ≥ 2.2 - <5] versus high levels, [MET/CEP7 ratio > 5]).

218 Statistical analysis

219 The METROS was a phase II, two arms, non-comparative trial in which arms were
220 determined by the presence of *ROS1* rearrangement or *MET* deregulation. The study was
221 designed to test the hypothesis of an ORR $\geq 50\%$ versus ORR $\leq 10\%$ in each arm at a
222 significance level of 5% (one sided) with a power of 98%. The study was originally
223 designed to include only *MET* amplified NSCLCs. However, clinical data published in 2015
224 suggested *MET* exon 14 mutated NSCLCs as an additional population potentially
225 benefiting to crizotinib (8,21). For such reason the trial was amended to include also
226 patients with such aberration without modification in the statistical plan. Patients and
227 disease characteristics were analyzed using descriptive statistics and expressed as
228 relative frequency (percentage) for discrete variables or median and interquartile range
229 (IQR) for continuous variables. Associations among factors were evaluated with the chi-
230 square test while differences in distribution of quantitative variables were measured with
231 the Mann-Whitney test. Confidence interval (95%) for ORR was calculated according to
232 the exact method. PFS and OS were calculated from the date of starting therapy to the
233 date of first evidence of either disease progression or death of the patient in the absence
234 of documented disease progression (PFS), or death for any cause (OS). Patients without
235 an event were censored at the date of last follow-up. Survival times were estimated using
236 Kaplan–Meier analysis and expressed as medians with corresponding two-sided 95%

237 confidence intervals (CIs). Differences between curves were evaluated using the log-rank
238 test. This trial is registered with ClinicalTrials.gov, number NCT 02499614.

239

240 Results

241 Patients

242 From December 2014 to March 2017, 505 patients were screened (Fig. 1). Four hundred
243 and thirty-three (86%) patients had tumor tissue evaluable for *ROS1* and *MET* FISH
244 analyses. Thirty-three individuals (7.6%) resulted *ROS1* positive and 37 *MET* deregulated
245 (8.5%). Among them, 18 patients were not included due to death (3 *ROS1* and 4 *MET*
246 patients), screening failure (4 *ROS1* and 4 *MET* patients) or unknown reason (3 *MET*
247 patients). Twenty-six patients per cohort (cohort A, *ROS1* positive; cohort B, *MET* positive)
248 accounted for the final population of the trial. Demographic and disease characteristics
249 are reported in table 1. Briefly, cohort A included mainly females, never smokers and with
250 median age of 68 years, whereas cohort B included mainly males, current or past smokers
251 with median age of 56 years. In both cohorts, most patients had an ECOG PS of 0-1,
252 presented with two or more metastatic sites, and received crizotinib as second line
253 treatment. Fifty-four percent of patients in the *MET* cohort had PD as best response to last
254 therapy compared to less than 30% in *ROS1* cohort. A platinum-doublet regimen was the
255 latest treatment before crizotinib in 69% and 81% of *ROS1* and *MET* deregulated patients,
256 respectively (Supplemental Table S1). Among individuals included in the *MET* cohort, 16
257 patients had *MET* amplification (intermediate levels, 14 patients; high levels, 2 patients), 9
258 patients had exon 14 mutation (c.2962C>T, 1 case; c.3029C>T, 5 cases; c.3082G>T, 1
259 case; c.3082+1G>A, 1 case; c.3082+3A>T, 1 case), and one patient had co-occurrence of
260 *MET* amplification (intermediate levels) and *MET* mutation (c.2942-19_2961delinsC). At
261 the end of the study, *MET* mutational status was centrally re-tested in all exon 14 mutant
262 cases, with confirmation of local reports. Only one case resulted unconfirmed due to

263 inadequate tumor sample for direct sequencing (data not shown). This patient had stable
264 disease as best response to study drug.

265 Median number of crizotinib cycles and median duration of treatment were 15 (range 0.3 -
266 34.4; IQR 5.4-24.9) and 15.2 months (95% CI 4.7-25.2) in cohort A, and 3 (range 0.4-28.6;
267 IQR 2.0 – 5.6) and 4.0 months (95% CI 2.0-5.5) in cohort B. At data cutoff, 9 patients in
268 cohort A and 6 patients in cohort B were still receiving treatment.

269 *Efficacy*

270 Summary of efficacy measures is reported in table 2. In Cohort A, ORR was 65% (CI 44 –
271 82), including one (4%) CR and 16 (61%) PR. Six patients (23%) obtained SD, resulting in
272 an overall disease control rate (DCR) of 85%. With a median follow-up of 21 months (95%
273 CI 19.0 – 24.5), median PFS was 22.8 months (95% CI 15.2 – 30.3), whereas median OS
274 was not reached (Fig. 2,3). Median time to response (TTR) and median duration of
275 response (DOR) were 7.9 weeks (IQR, 7.4-10.3) and 21.4 months (95% CI 12.7-30.1).

276 Depth of response, defined as the median percentage of reduction in target lesions from
277 baseline, was -51.7% (IQR, -77.5% - -42.7%). In responding patients, median percentage
278 of *ROS1* FISH positivity was significantly higher than in non-responders (50% versus 22%,
279 Mann-Whitney test $p=0.005$, Supplemental Table S2).

280 In cohort B, ORR was 27% (CI 11 – 47), including only PR. Eleven patients (42%) had SD,
281 for an overall DCR of 69%. With a median follow-up of 21 months (95% CI 19.0 – 24.5),
282 median PFS and median OS were 4.4 months (95% CI 3.0 – 5.8) and 5.4 months (95% CI
283 4.2– 6.5), respectively (Fig. 2,3). Median TTR and DOR were 7.4 weeks (IQR 6.4-9.3) and
284 3.7 months (95% CI 1.1-6.3). Depth of response was -47.9% (-56.5%- -35.6%). According
285 to *MET* deregulation, responses occurred in 5 patients with *MET* amplification
286 (intermediate levels only), in 1 patient with *MET* mutation and in the co-altered *MET*

287 amplified and mutant case (Supplemental Table S3). Further, we separately analyzed
288 outcome in *MET* amplified and in *MET* exon 14 mutated groups. In both groups ORR,

289 median DOR, PFS, and OS were similar to the whole *MET* deregulated cohort
290 (Supplemental Table S4).

291 Further, to better characterize the study population, we retrospectively performed a
292 Sequenom analysis in 48 out of 52 tissue specimens collected at baseline. A second
293 driver was found in 7 (14%) cases, including two cases with concomitant *MET*
294 amplification (intermediate levels), three cases with *MET* exon 14 mutation and two cases
295 with *ROS1* rearrangement. Details are reported in supplemental Tables S2-S3. Among the
296 4 evaluable patients with *MET/KRAS* and *ROS1/KRAS* co-altered tumors, 1 achieved PR,
297 2 had SD, and 1 progressed. The double *MET* amplified/*BRAF* mutant subject progressed,
298 whereas the *ROS1/MET* positive patient voluntarily discontinued crizotinib after only 1 cycle
299 without any tumor assessment.

300 Finally, as illustrated in Table 3, we evaluated the intracranial efficacy of the drug in the
301 eleven patients with brain metastases at baseline (six in Cohort A and five in Cohort B)
302 and responses were observed only in *ROS1* positive patients.

303

304 *Toxicity*

305 Safety profile of crizotinib was consistent with literature data and no new safety alert was
306 reported in both cohorts (Supplemental Table S5). Treatment related adverse events
307 (TRAEs), most of which were of grade 1 or 2, occurred in 26 (100%) patients in Cohort A
308 and in 21 (81%) patients of cohort B. In both cohorts, the most common TRAEs of any
309 grade were fatigue (58% in cohort A and 31% in cohort B), peripheral edema (50% in
310 cohort A and 31% in cohort B), nausea (46% in cohort A and 31% in cohort B), pain (30%
311 in cohort A and 19% in cohort B), transaminases elevation (27% in both cohort A and
312 cohort B), respiratory symptoms including dyspnea and cough (42% in cohort A and 46%
313 in cohort B), and visual disorders (23% in cohort A and 27% in cohort B). In cohort A,
314 TRAEs of grade 3/4 were peripheral edema, neutropenia and respiratory symptoms each

315 occurring in one patient (4%), and nausea and fatigue each occurring in two (8%) patients.
316 In cohort B, TRAEs of grade 3/4 were nausea, neutropenia, anemia, and respiratory
317 symptoms each occurring in one patient (4%), nausea and transaminases elevation
318 occurring in two patients (8%). Overall, TRAEs leading to dose reduction, temporary or
319 permanent of discontinuation of the drug were reported in 8 (15%), 13 (25%), and 3 (6%)
320 patients. Among 13 serious AEs (SAEs) reported, only two were judged as related to study
321 drug (Supplemental Table S6). Finally, we analyzed the incidence and clinical correlates of
322 venous thromboembolism occurring prior or during crizotinib treatment in patients enrolled
323 in the trial; the results of this analysis are the object of a separate publication (26).
324

325 Discussion

326 Treatment of patients with *MET* deregulated NSCLC represents an urgent clinical need
327 because of lack of effective targeted therapies and unfavorable prognosis (7,27). The
328 METROS is a prospective study evaluating the efficacy of crizotinib in patients with *MET*
329 exon 14 mutations or amplification. Although response rate was remarkable for a
330 pretreated NSCLC population, median PFS and, most importantly, median OS were
331 disappointing, with all patients rapidly progressing and dying.
332 In oncogene addicted NSCLCs, such as *EGFR* or *ALK* addicted NSCLC, targeted
333 therapies are extending survival with medians ranging between 3-5 years.(1–5) Similar
334 outcome has been observed in *ROS1* addicted patients and the results of our study,
335 including also a *ROS1* cohort, confirmed that crizotinib is highly effective in such patients.
336 The primary end-point of ORR was met in *ROS1* cohort, where durable responses were
337 observed in 65% of patients, median PFS exceeded 22 months and median OS was not
338 reached, with approximately 80% of patients alive at 1 year. These data favorable
339 compare with other trials, reinforcing the role of this agent in the treatment of *ROS1* driven
340 lung cancers (10–14). Interestingly, we observed a significant association between

percentage of *ROS1* FISH positivity and response to crizotinib, a phenomenon previously described only in *ALK* positive NSCLC, deserving further investigations (28).

Different results were obtained in *MET* deregulated patients. This cohort included both *MET* amplified or exon 14 mutated. Although recent data suggest that these are different patient population (19,23), this concept was not emerged yet at the time of trial design and statistical hypothesis has been formulated considering *MET* deregulated as a homogeneous group. However, even with such limitation, outcome was similar in *MET* amplified or exon 14 mutated subgroups, with limited responses and with only one month elapsing from time of tumor progression and patient death. These data are in agreement with other studies, such as the AcSé and the PROFILE 1001 (29, 30). Final results of the AcSé *MET* FISH positive cohort, showed an ORR of 32% and a median PFS of only 3.4 months, comparable to what observed in our experience, even if criteria for *MET* positivity differed (29). In the last update of the PROFILE 1001, including a total of 37 *MET* amplified patients, ORR was 27%, similar to the 31% observed in METROS. Importantly, among the 20 patients with high levels of amplification ORR was 40%, including 2 cases with CR (30). In our study, only one patient had high levels of amplification, precluding the possibility to explore the impact of the drug in presence of such characteristic. Nevertheless, high levels of *MET* amplification rarely occur in patients with NSCLC. In a previous study conducted in surgically resected NSCLC, among 435-screened patients only three (0.6%) had high levels of amplification (27). METROS trial screened more than 430 advanced NSCLCs and only 0.4% displayed high levels of *MET* amplification, confirming the relative rarity of the event.

METROS study also included patients with *MET* exon 14 mutations, accounting for less than 3% of the screened population, as expected according to literature data (7,8). In *MET* exon 14 mutated patients the benefit produced by crizotinib in terms of ORR, PFS and OS was limited. Although our findings seem inferior to what recently reported by Drillon et al.

367 in PROFILE 1001, in which PFS exceed 7 months and OS is approximately 20 months,
368 differences in patients selection limit comparison between the two studies (30). Exon 14
369 mutations include a wide range of abnormalities, such as insertion, deletion or point
370 mutation that generally lead the loss or attenuation of ubiquitin-mediated receptor
371 degradation, for instance by disrupting the splice acceptor site of intron 13 or affecting the
372 splice donor site of intron 14 (8,27,32). How different mutations could affect sensitivity to
373 MET inhibitors, especially crizotinib, remains an unanswered question. In our cohort
374 among the 4 patients with splicing mutations only one responded.

375 METROS study also confirms the very unfavorable prognosis of *MET* deregulated NSCLC.
376 In 2009 our group first demonstrated that *MET* gene copy number was a negative
377 prognostic factor in NSCLC (27). Additional studies confirmed that *MET* deregulation,
378 including overexpression, gene copy number gain or mutation, confers an aggressive
379 phenotype (33). In addition to an aggressive behavior, also reinforced by the scarce
380 sensitivity to prior chemotherapy, *MET* deregulated NSCLC demonstrated modest and
381 transient responsiveness to crizotinib, suggesting that other factors could modulate
382 sensitivity to *MET*-inhibition, such as co-occurrence of driver events or expression of the
383 *MET* protein as recently reported (33–36). Particularly, in a context of *MET* amplification
384 Tong et al. demonstrated that low levels of amplification may occur in a background of
385 *KRAS* mutation, while high levels of *MET* amplification were mutually exclusive with major
386 oncogene drivers (33). In addition, it is not possible to exclude that other approaches or
387 new drugs might be more effective. On this perspective, we are now conducting a phase II
388 trial evaluating cabozantinib in both *MET* amplified or mutated lung cancer untreated with
389 *MET*-inhibitor or refractory to crizotinib (CABinMET trial, EudraCT 2017-004157-16).

390 In conclusion, results of METROS trial indicate that even if crizotinib induces a tumor
391 shrinkage in a fraction of *MET* deregulated NSCLC, the drug minimally impacts the clinical
392 course of the disease, at least in pretreated *MET* mutated or *MET* with intermediate levels

393 of amplification, whereas the efficacy of the drug in presence of high levels of amplification
394 remains investigational. Additional studies and innovative therapies are urgently needed
395 against this aggressive disease.

396

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Table 1. Baseline Characteristics

	Cohort A ROS1 ^{FISH+}		Cohort B MET ^{Ex14} or MET ^{FISH+}		p*
	26	100%	26	100%	
Age, median (range)	68 (28-86)		56 (39-78)		0.07
M/F	10/16	38/62	17/9	65/35	0.05
ECOG PS 0/1/2	18/7/1	69/27/4	11/13/2	42/50/8	0.05
Never Smoker/Past Smoker/Current smoker	14/9/3	54/35/11	6/12/8	23/46/31	0.05
Adenocarcinoma/Other histology**	26/0	100/0	23/3	89/11	n.a.
Type of MET deregulation					
• Amplification	n.a.		16	61	
• Mutation	n.a.		9	35	
• Concomitant amplification and mutation	n.a.		1	4	
Metastatic sites, 1/2/>2	5/11/10	19/42/39	4/11/11	15/42/42	0.92
Disease sites					
• Lung	22	85	22	85	0.99
• Lymph Node	16	61	12	46	0.40
• Liver	3	11	6	23	0.46
• Bone	9	35	5	19	0.35
• Brain	6	23	5	19	0.73
• Pleura	3	11	6	23	0.46
• Adrenal gland	2	7	6	23	0.25
Prior line of therapy, 1/2/>2	20/3/3	76/12/12	21/3/2	81/11/8	0.63
Time from end of last treatment to Crizotinib start (months, median)	12 (3-43)		3 (2-8)		0.02
Best response to prior therapy					
• Complete Response + Partial Response	10	38	4¶	15	0.12
• Stable Disease	7	27	6	23	
• Progressive Disease	7	27	14	54	
• Unknown	2	8	2	8	

Data are median (IQR) or n (%). * Chi-square test and Mann-Whitney were used for categorical items and for continuous variables, respectively. ECOG PS=Eastern Cooperative Oncology Group performance status.**Other histologies includes two patients with NSCLC not otherwise specified (NOS) and one patient with pleomorphic carcinoma. All histology was determined by local pathological report.n.a.= not applicable; ¶ only partial responses.

Table 2. Efficacy endpoints in cohort A (ROS1 positive) and Cohort B (MET deregulated)

	Cohort A – ROS1 (N=26)	Cohort B– MET (N=26)
Best overall response		
Complete response	1 (4%)	0
Partial response	16 (61%)	7 (27%)
Stable disease	6 (23%)	11 (42%)
Progressive disease	1 (4%)	6 (23%)
Not evaluable or not assessed	2 (8%)	2 (8%)
Progression free survival (PFS)		
Number of events	14 (54%)	18 (69%)
PFS (months)	22.8 (15.2 – 30.3)	4.4 (3.0- 5.8)
6 months PFS rate	80.6%	30.9%
12 months PFS rate	71.9%	20.6%
Overall survival (OS)		
Number of events	10 (39%)	16 (61%)
OS (months)	NR	5.4 (4.2 – 6.5)
6 months OS rate	96.2%	43.9%
12 months OS rate	79.2%	26.3%

Data are n (%) or median (95% CI), unless otherwise stated. NR, not reached.

536 **Table 3. Intracranial disease characteristics, intracranial response and pattern of failure in patients with brain metastases at**
537 **baseline**

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Pt ID	Biomarker	Characteristics of CNS disease	Method for CNS Assessment	Prior RT	Date of RT	Date of start crizotinib	Intracranial response	Pattern of failure
MT-006-118	<i>ROS1</i>	Multiple lesions, non-target	Brain MRI	No	NA	01 Dec 2015	CR	Extra- and intracranial
MT-001-013	<i>ROS1</i>	Single lesion, non-target	Brain CT scan	No	NA	31 Mar 2015	SD	Intracranial only
MT-001-001	<i>ROS1</i>	Single lesion, non-target	Brain CT scan	Yes	Jun 2012	27 Feb 2015	CR	Extra- and intracranial
MT-012-087	<i>ROS1</i>	Single lesion, non-target	Brain CT scan	No	NA	27 Jan 2016	SD	Extracranial only
MT-001-011	<i>ROS1</i>	Single lesion, non-target	Brain CT Scan	No	NA	20 Feb 2015	SD	Intracranial only
MT-019-238	<i>ROS1</i>	Multiple lesions, non-target	Brain MRI	Yes	NR	25 Jul 2016	SD	Intracranial only
MT-006-079	<i>MET</i> ^{FISH+}	Single lesion, non- target	Brain MRI	Yes	Aug 2015	10 Aug 2015	PD	Extra- and intracranial
MT-012-182	<i>MET</i> ^{FISH+}	Multiple lesions, target (1 lesion)	Brain CT scan	Yes	Apr 2016	26 Jul 2016	SD	Extracranial only
MT-004-286	<i>MET</i> ^{Ex14}	Multiple lesions, non-target	Brain MRI	Yes	Jun 2016	14 Sep 2016	SD	NA§
MT-012-129	<i>MET</i> ^{Ex14}	Multiple lesions, target (1 lesion)	Brain CT scan	No	NA	22 Sep 2016	PD	Extra- and intracranial
MT-020-441	<i>MET</i> ^{Ex14}	Single lesion, non-target	Brain CT scan	Yes	Jun 2017	05 Jul 2017	SD	Extracranial only

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540 *MET*^{FISH+}, *MET* amplified (intermediate level only, *MET*/*CEP7* ratio 3.4 and 2.6 for MT-006-079 and MT-012-182, respectively); RT, radiotherapy;

541 NR, not reported; NA, not applicable; § PT ID MT-004-286 permanently discontinued crizotinib due to SAE

Figure legend

Figure 1. Study profile.

Figure 2. Tumor responses in *MET* deregulated and *ROS1* rearranged NSCLC.

Maximum percentage reduction from baseline sum of lesion diameters by best investigator-assessed confirmed response in 52 patients receiving crizotinib as second-line or later treatment.

Figure 3. Kaplan-Meier curve of investigator-assessed progression-free survival (A) and overall survival (B) in *ROS1* positive (red line) and *MET* deregulated (blue line) non-small cell lung cancer receiving crizotinib.

Figure

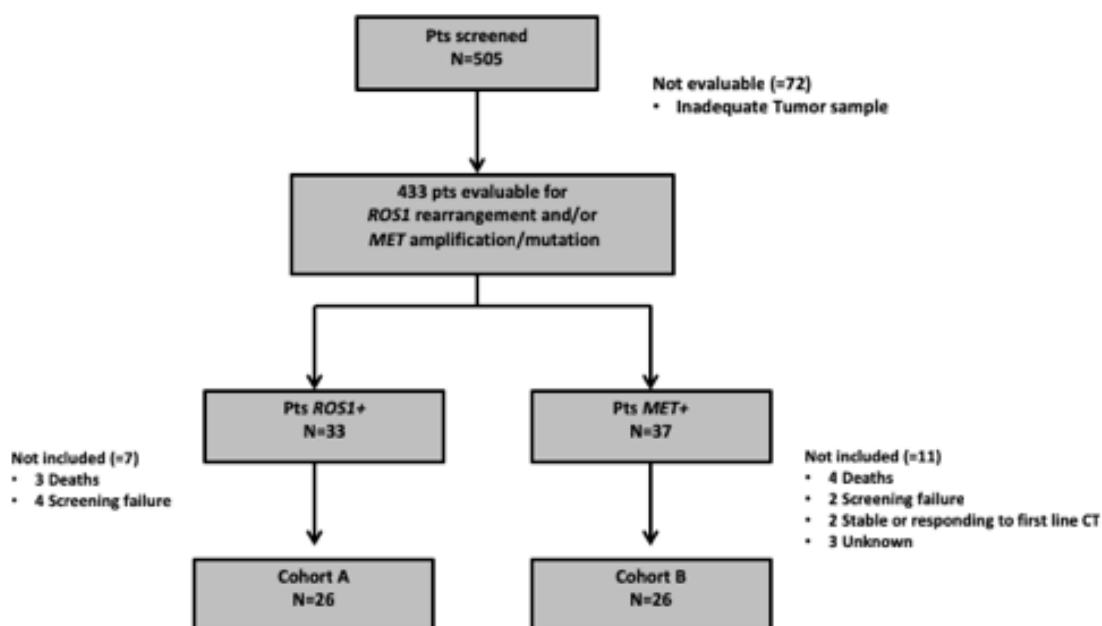


Figure 2

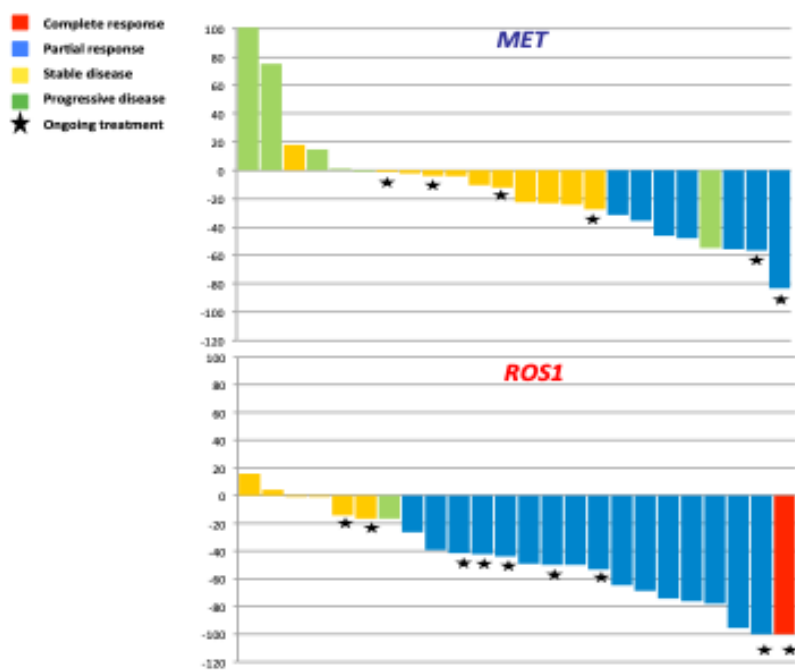


Figure 3

